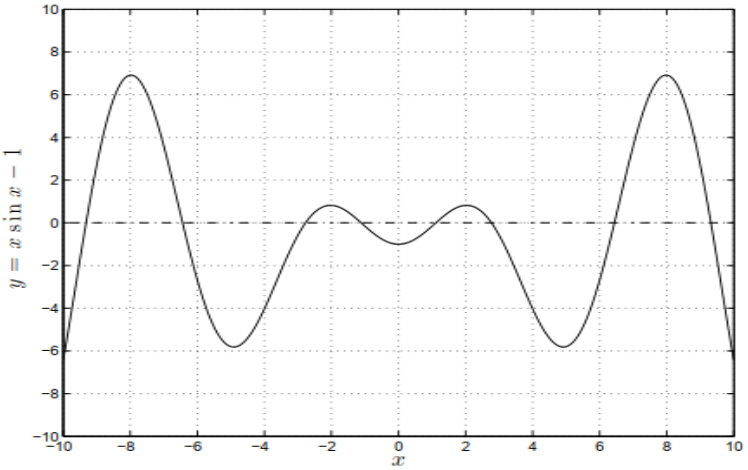
Name: Enrolment No:	

UPES

Assignment-1

Programme Name : B. Tech. SoCS/SoAE **Semester : II**
Course Name : Advanced Engineering Mathematics-II **Max. Marks: 100**
Course Code : MATH1065 **Unit(s) : I & II**

Date of Submission: 25/02/2026

S. No.		Marks	CO
1	A chemist is preparing three mixtures using chemicals A, B, and C. In the first mixture, the required quantities are 10 parts of A, 1 part of B, and 1 part of C, giving a total weight of 4.2 units. In the second mixture, the required quantities are 2 parts of A, 10 parts of B, and 1 part of C, giving a total weight of 6.3 units. In the third mixture, the required quantities are 2 parts of A, 2 parts of B, and 10 parts of C, giving a total weight of 8.6 units. Formulate and solve the resulting system of equations using Gauss- Seidal method to determine the amounts of A, B, and C.	**10**	**CO1**
2	Consider the nonlinear equation $x \sin x - 1 = 0$. (i) Find an initial guess x_0 such that $x_0 > 1$, with the help of the graph depicted in Figure below, such that the Newton-Raphson method is likely to converge to the solution x^* of the given nonlinear equation lying in the interval $(-10, -9)$. Compute x_1, x_2, x_3, x_4 of the corresponding Newton-Raphson iterative sequence. (ii) Find another initial guess x_0 such that $x_0 > 1$, with the help of the graph, such that the Newton-Raphson method is likely to converge to the smallest positive solution of the given nonlinear equation.  **Figure: Graph of $x\ sin\ x - 1$**	**10**	**CO1**

3	The velocity of a falling object is given as $v(t) = \sqrt{\frac{gm}{c_d}}\tanh\left(\sqrt{\frac{gc_d}{m}}t\right)$ where the upward force due to air resistance is assumed to be proportional to the square of the velocity. Here, c_d denotes the second-order drag coefficient. (a) If $g = 9.81\,m/s^2$, $m = 68.1\,kg$, and $c_p = 0.25\,kg/m$, use analytical integration to determine how far the object falls in $10\,s$. (b) Compute the same distance by numerically evaluating the integral using the trapezoidal rule. Choose a sufficiently large number of subintervals $n$so that the result is accurate to three significant digits.	**10**	**CO1**
4	Solve the following problem over the interval from $x = 0$ to 1 using a step size of 0.25 where $y(0) = 1$. $$\frac{dy}{dt} = (1 + 4t)\sqrt{y}$$ Use, (a) Analytical method, (b) Euler's method and (c) Fourth-order RK method. Display all your results on the same graph. Thus, comment on the errors obtained by distinct numerical methods compared to the exact analytical solution.	**10**	**CO1**
5	To infer the surface shape of an object from images taken of a surface from three different directions, one needs to solve the following set of equations. $$\begin{bmatrix} 0.2425 & 0 & -0.9701 \\ 0 & 0.2425 & -0.9701 \\ -0.2357 & -0.2357 & -0.9428 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 247 \\ 248 \\ 239 \end{bmatrix}$$ The right hand side values are the light intensities from the middle of the images, while the coefficient matrix is dependent on the light source directions with respect to the camera. The unknowns are the incident intensities that will determine the shape of the object. Find the values of x_1, x_2 and x_3 using the Gauss-Seidel method. Use $$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix}$$ as the initial guess and conduct two iterations.	**10**	**CO1**
6	A construction company undertakes the building of residential houses and commercial units using two specialized work crews. Crew A, responsible for structural work, requires 5 man-days to construct one commercial unit and 2 man-days to construct one residential house. Crew B, responsible for finishing work, requires 3 man-days for completing either type of building. Due to labor and equipment constraints, Crew A is available for 180 man-days per week and Crew B is available for 135 man-days per week.	**10**	**CO1**

	The company earns a profit of Rs. 300 from each commercial unit and Rs. 200 from each residential house. How many residential houses and commercial units should the company construct per week in order to maximize total profit? Formulate the problem as a linear programming problem and solve it using the Simplex method.		
7	In microchip design, electric potential distribution in a thin conducting plate is modeled using harmonic functions. Suppose the electric potential in a 2D region is: $$u(x, y) = x^2 - y^2 + 2x$$ 1) Show that $u(x, y)$is harmonic. 2) Find its harmonic conjugate $v(x, y)$. 3) Hence find the analytic function $f(z)$. 4) Explain what this means physically in chip design.	**10**	**CO2**
8	Answer the following in brief: 1) What does it mean for a function to be analytic at a point? 2) Why must a function satisfy the Cauchy–Riemann equations to be analytic? 3) Why are analytic functions considered "smooth" in engineering analysis?	**15**	**CO2**
9	Consider the following functions: $$f_1(z) = z^2, f_2(z) = \mid z \mid^2, f_3(z) = \bar{z}$$ 1) Identify which functions are analytic. 2) Explain **why** some functions fail to be analytic.	**15**	**CO2**